

JEE MAIN REVISION MASTER: INDEFINITE INTEGRATION 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Indefinite Integration - Core Application	28 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Indefinite Integration - Core Application (Very High Yield — 28 PYQs)

Core Principle: Key problem pattern in Kcet Indefinite Integration.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int e^{-x} \log 2 \cdot 2^x dx = \dots$ " [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If 'n' is a natural number, then $\int \frac{\sin^n x \cos^{n+2} x}{\cos^2 x} dx = \dots$ " [Ans: D]
- [2025 Shift] Q#3 (MCQ): "The value of $\int \frac{dx}{(x+1)(x+2)}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\int e^{-x} \log 2 \cdot 2^x dx =$$

- (A) $\log x + C$
- (B) $x + C$ [Correct]
- (C) $(1/x) + C$
- (D) $\frac{x^2}{2} + C$

Correct Answer: B

Solution Analysis: Since 2^x can be written as $2^x = e^{x \log 2}$ the integrand becomes $e^{-x} \log 2 \cdot 2^x = e^{-x} \log 2 \cdot e^{x \log 2}$. Using $e^a \cdot e^b = e^{a+b}$, $e^{-x} \log 2 + x \log 2 = e^0 = 1$. So the integral is $\int 1 dx = x + C$. So the correct option is $x + C$ Option B.